



## General Chemistry, First-Term Practice Items

### Study Question – Chapter 0 (Toolbox)

- 1A.** A 15.75 g piece of iron is carefully placed into a graduated cylinder containing 10.0 mL of water. What is the final reading on the graduated cylinder and how will a 15.75 g piece of copper raise the water in comparison?

| density / $\text{g}\cdot\text{mL}^{-1}$ |      |
|-----------------------------------------|------|
| iron                                    | 7.87 |
| copper                                  | 8.92 |

- (A) Final volume of 10.5 mL, and copper will raise the water less than iron.  
(B) Final volume of 10.5 mL, and copper will raise the water more than iron.  
(C) Final volume of 12.0 mL, and copper will raise the water less than iron.  
(D) Final volume of 12.0 mL, and copper will raise the water more than iron.

Work:

### Practice Question – Chapter 0 (Toolbox)

- 1B.** Which compound is named correctly?
- (A)  $\text{SO}_3$ ; sulfite ion  
(B)  $\text{FePO}_4$ ; iron(II) phosphate  
(C)  $\text{H}_2\text{S}(\text{aq})$ ; hydrosulfuric acid  
(D)  $\text{CuSO}_4\cdot 5\text{H}_2\text{O}$ ; copper sulfate pentahydrate

Work:

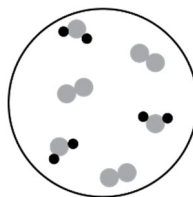
*References below are to the ACS Exams General Chemistry Study Guide*

For more practice, please see: Study Questions: SQ3-6 (p. 1-5)  
Practice Questions: PQ5-12, PQ-13 (p. 6-7)

Study Question – Chapter 0 (Toolbox)

**2A.** What is the classification of the diagram shown?

- (A) a pure compound
- (B) a mixture of two elements
- (C) a mixture of two compounds
- (D) a mixture of one compound and one element



Work:

Practice Question – Chapter 0 (Toolbox)

**2B.** What is the longest length?

- (A)  $10^{-5}$  km
- (B) 1 m
- (C) 100 mm
- (D)  $10^7$  nm

Work:

For more practice, please see: Study Questions: SQ1, 7 (p. 1-5)  
Practice Questions: PQ1-2, 14-15 (p. 6-7)

## Study Question – Chapter 1 (Atomic Structure)

**3A.** How many protons, neutrons and electrons are in one atom of aluminum-27 ( $^{27}\text{Al}$ )?

|     | protons | neutrons | electrons |
|-----|---------|----------|-----------|
| (A) | 13      | 13       | 13        |
| (B) | 13      | 14       | 13        |
| (C) | 14      | 13       | 14        |
| (D) | 13      | 27       | 13        |

Work:

## Practice Question – Chapter 1 (Atomic Structure)

**3B.** How many subatomic particles are in one **ion** of oxygen-18 ( $^{18}\text{O}^{2-}$ )?

|     | protons | neutrons | electrons |
|-----|---------|----------|-----------|
| (A) | 8       | 8        | 6         |
| (B) | 8       | 10       | 8         |
| (C) | 8       | 10       | 10        |
| (D) | 8       | 18       | 8         |

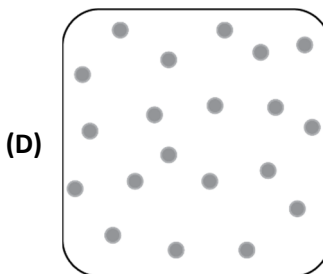
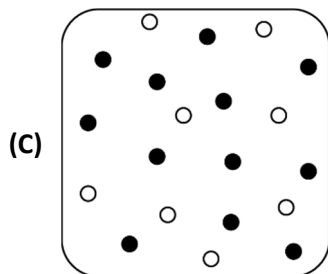
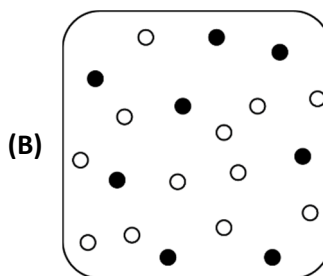
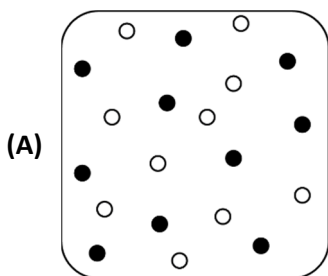
Work:

For more practice, please see: Study Questions: SQ2-4, (p. 16-19)  
Practice Questions: PQ4-13 (p. 20-22)

Study Question – Chapter 1 (Atomic Structure)

**4A.** There are two stable isotopes of gallium ( $Z = 31$ ), gallium-69 and gallium-71. Which figure best represents a sample of gallium?

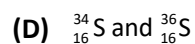
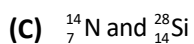
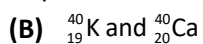
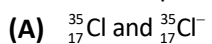
○ = gallium-69  
● = gallium-71  
● = gallium-69.72



Work:

Practice Question – Chapter 1 (Atomic Structure)

**4B.** Which set represents isotopes of the same element?

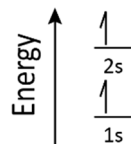


Work:

For more practice, please see: Study Questions: SQ1, 5-6 (p. 16-19)  
Practice Questions: PQ1-3, 14-17 (p. 20-22)

Study Question – Chapter 2 (Electronic Structure)

**5A.** For the orbital diagram of helium shown, what is a possible set of quantum numbers of the excited electron?



**(A)** (1, 0, 0,  $\frac{1}{2}$ )

**(B)** (1, 1, 0,  $\frac{1}{2}$ )

**(C)** (2, 0, 0,  $\frac{1}{2}$ )

**(D)** (2, 1, 0,  $\frac{1}{2}$ )

Work:

Practice Question – Chapter 2 (Electronic Structure)

**5B.** What is the ground state electron configuration for tin (Sn,  $Z = 50$ )?

**(A)** [Kr]  $4s^2 3d^{10} 4p^2$

**(B)** [Kr]  $5s^2 4d^{10} 5p^2$

**(C)** [Kr]  $5s^2 5d^{10} 5p^2$

**(D)** [Kr]  $5s^2 5p^2$

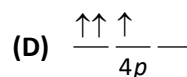
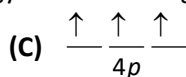
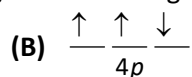
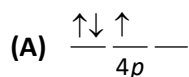
Work:

For more practice, please see: Study Questions: SQ3-6 (p. 26-31)  
Practice Questions: PQ7-13 (p. 32-34)



Study Question – Chapter 2 (Electronic Structure)

**6A.** What is the orbital diagram for the highest energy sublevel of a ground state atom of arsenic (As)?



Work:

Practice Question – Chapter 2 (Electronic Structure)

**6B.** Which element has the highest first ionization energy?

(A) As

(B) Ge

(C) Sb

(D) Sn

Work:

For more practice, please see: Study Questions: SQ7-11 (p. 26-31)  
Practice Questions: PQ14-30 (p. 32-34)



Study Question – Chapter 3 (Formula Calculations and the Mole)

7A. Which sample contains the greatest number of atoms?

(A) 1.0 g H<sub>2</sub>O

(B) 1.0 g H<sub>2</sub>O<sub>2</sub>

(C) 2.0 g H<sub>2</sub>O

(D) 2.0 g H<sub>2</sub>O<sub>2</sub>

Work:

Practice Question – Chapter 3 (Formula Calculations and the Mole)

7B. What is the mass (in g) of  $3.75 \times 10^{22}$  molecules of methanol, CH<sub>3</sub>OH?

| molar mass / g·mol <sup>-1</sup> |       |
|----------------------------------|-------|
| CH <sub>3</sub> OH               | 32.05 |

(A) 0.0623 g

(B) 0.501 g

(C) 2.00 g

(D) 16.1 g

Work:

For more practice, please see: Study Questions: SQ1-5 (p. 37-42)  
Practice Questions: PQ1-18 (p. 42-44)



Study Question – Chapter 3 (Formula Calculations and the Mole)

**8A.** What is the empirical formula for a hydrocarbon containing 83.6% carbon and 16.4% hydrogen by mass?

**(A)**  $\text{CH}_2$

**(B)**  $\text{CH}_{2.33}$

**(C)**  $\text{C}_3\text{H}_7$

**(D)**  $\text{C}_5\text{H}$

Work:

Practice Question – Chapter 3 (Formula Calculations and the Mole)

**8B.** Which compound has the greatest percent composition by mass of chlorine?

**(A)**  $\text{KClO}_3$

**(B)**  $\text{KClO}_4$

**(C)**  $\text{NaClO}_3$

**(D)**  $\text{NaClO}_4$

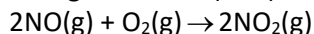
Work:

For more practice, please see: Study Questions: SQ6-8 (p. 37-42)  
Practice Questions: PQ19-30 (p. 42-44)

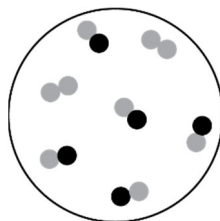


Study Question – Chapter 4 (Stoichiometry)

- 9A.** Nitrogen monoxide (NO) can react with oxygen to produce nitrogen dioxide (NO<sub>2</sub>):



Given the reaction vessel shown, how many moles of NO<sub>2</sub> can be made if each molecule represents 1.0 mole?

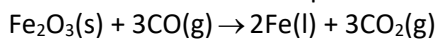


- (A) 2.0 mol NO<sub>2</sub>      (B) 2.5 mol NO<sub>2</sub>      (C) 4.0 mol NO<sub>2</sub>      (D) 5.0 mol NO<sub>2</sub>

Work:

Practice Question – Chapter 4 (Stoichiometry)

- 9B.** Iron(III) oxide can react with carbon monoxide to produce liquid iron and carbon dioxide:



What mass (in g) of carbon monoxide is needed to produce 275 g of iron?

| molar mass / g·mol <sup>-1</sup> |       |
|----------------------------------|-------|
| CO                               | 28.01 |
| Fe                               | 55.85 |

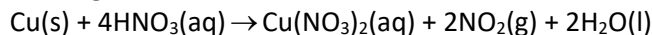
- (A) 69.0 g      (B) 91.9 g      (C) 207 g      (D) 414 g

Work:

For more practice, please see: Study Questions: SQ3-4 (p. 48-51)  
Practice Questions: PQ7-18 (p. 52-55)

Study Question – Chapter 4 (Stoichiometry)

- 10A.** Copper will react with nitric acid ( $\text{HNO}_3$ ) to produce copper(II) nitrate ( $\text{Cu}(\text{NO}_3)_2$ ), nitrogen dioxide ( $\text{NO}_2$ ) and water:



| molar mass / $\text{g}\cdot\text{mol}^{-1}$ |       |
|---------------------------------------------|-------|
| Cu                                          | 63.55 |
| $\text{NO}_2$                               | 46.01 |

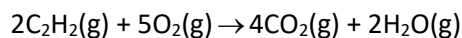
Suppose  $\text{NO}_2$  is produced at a percent yield of 85%. What mass (in g) of copper needs to be used in an excess of  $\text{HNO}_3$  to experimentally produce 10.0 g of  $\text{NO}_2$ ?

- (A) 5.9 g Cu      (B) 8.1 g Cu      (C) 12 g Cu      (D) 16 g Cu

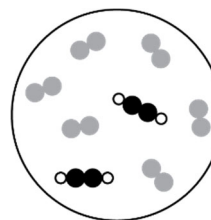
Work:

Practice Question – Chapter 4 (Stoichiometry)

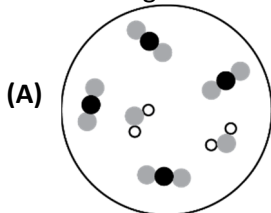
- 10B.** Acetylene ( $\text{C}_2\text{H}_2$ ) reacts with oxygen to produce carbon dioxide and water:



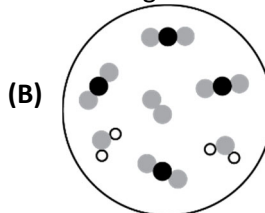
Given the diagram to the right, what is the limiting reactant and what does the reaction vessel look like when the reaction is over (assuming 100% yield)?



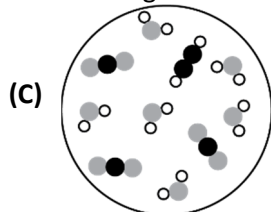
Limiting reactant =  $\text{C}_2\text{H}_2$



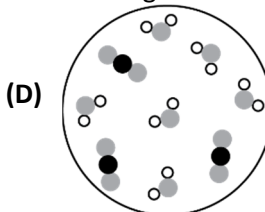
Limiting reactant =  $\text{C}_2\text{H}_2$



Limiting reactant =  $\text{O}_2$



Limiting reactant =  $\text{O}_2$



Work:

For more practice, please see: Study Questions: SQ5-7 (p. 48-51)  
Practice Questions: PQ19-30 (p. 52-55)



Study Question – Chapter 5 (Solutions and Aqueous Reactions)

- 11A.** When iron(III) chloride ( $\text{FeCl}_3$ ) and sodium hydroxide ( $\text{NaOH}$ ) are mixed, the precipitate iron(III) hydroxide ( $\text{Fe}(\text{OH})_3$ ) forms. What are the spectator ions in this reaction?
- (A)  $\text{Fe}^{3+}$  and  $\text{Cl}^-$       (B)  $\text{Fe}^{3+}$  and  $\text{OH}^-$       (C)  $\text{Na}^+$  and  $\text{Cl}^-$       (D)  $\text{Na}^+$  and  $\text{OH}^-$

Work:

Practice Question – Chapter 5 (Solutions and Aqueous Reactions)

- 11B.** What is the oxidation number of nitrogen in  $\text{Fe}(\text{NO}_2)_3$ ?
- (A) +5      (B) +4      (C) +3      (D) -3

Work:

For more practice, please see: Study Questions: SQ5-11 (p. 58-63)  
Practice Questions: PQ14-30 (p. 64-67)



Study Question – Chapter 5 (Solutions and Aqueous Reactions)

**12A.** What mass (in g) of calcium chloride ( $\text{CaCl}_2$ ) is needed to make 125 mL of 0.100 M  $\text{Cl}^-$ ?

| molar mass / $\text{g}\cdot\text{mol}^{-1}$ |        |
|---------------------------------------------|--------|
| $\text{CaCl}_2$                             | 110.98 |

(A) 0.113 g

(B) 0.694 g

(C) 1.39 g

(D) 2.77 g

Work:

Practice Question – Chapter 5 (Solutions and Aqueous Reactions)

**12B.** What volume of water (in mL) must be added to a 1.25 M  $\text{HNO}_3$  solution to make 250. mL of a 0.575 M solution?

(A) 70. mL

(B) 115 mL

(C) 135 mL

(D) 180. mL

Work:

For more practice, please see: Study Questions: SQ1-4 (p. 58-63)  
Practice Questions: PQ1-10 (p. 64-67)



## Study Question – Chapter 5 (Solutions and Aqueous Reactions)

- 13A.** Aluminum nitrate ( $\text{Al}(\text{NO}_3)_3$ ) reacts with sodium sulfide ( $\text{Na}_2\text{S}$ ) to form a precipitate:  $2\text{Al}(\text{NO}_3)_3(\text{aq}) + 3\text{Na}_2\text{S}(\text{aq}) \rightarrow \text{Al}_2\text{S}_3(\text{s}) + 6\text{NaNO}_3(\text{aq})$

| molar mass / $\text{g}\cdot\text{mol}^{-1}$ |        |
|---------------------------------------------|--------|
| $\text{Al}_2\text{S}_3$                     | 150.17 |

When reacting 25.0 mL of the aluminum nitrate solution with an excess of sodium sulfide, 1.085 g of the precipitate is recovered at a 91.8% yield. What was the molar concentration of the aluminum nitrate solution?

- (A) 0.157 M      (B) 0.315 M      (C) 0.531 M      (D) 0.630 M

Work:

## Practice Question – Chapter 5 (Solutions and Aqueous Reactions)

- 13B.** During a titration, a strong acid ( $\text{H}_2\text{SO}_4$ ) can react with a strong base ( $\text{KOH}$ ) to produce water and a salt:  $\text{H}_2\text{SO}_4(\text{aq}) + 2\text{KOH}(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{K}_2\text{SO}_4(\text{aq})$

If 38.45 mL of a 0.193 M solution of  $\text{KOH}$  is needed to neutralize 25.0 mL of  $\text{H}_2\text{SO}_4$ , what is the molar concentration of the  $\text{H}_2\text{SO}_4$  solution?

- (A) 0.125 M      (B) 0.148 M      (C) 0.297 M      (D) 0.594 M

Work:

For more practice, please see: Study Questions: SQ1-4 (p. 58-63)  
Practice Questions: PQ11-13 (p. 64-67)

## Study Question – Chapter 6 (Heat and Enthalpy)

- 14A.** A 35 °C piece of copper with a mass of 25 g is placed into 25 mL of water at 25 °C. Both the metal and water reach the same final temperature. Assuming no heat is lost to the surroundings, what is true?

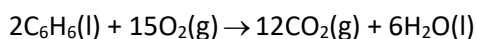
| specific heat / J·g <sup>-1</sup> ·°C <sup>-1</sup> |       |
|-----------------------------------------------------|-------|
| Cu                                                  | 0.385 |
| H <sub>2</sub> O                                    | 4.18  |

- (A)  $\Delta T_{\text{Cu}} > 0$       (B)  $\Delta T_{\text{water}} < 0$       (C)  $|\Delta T_{\text{Cu}}| > |\Delta T_{\text{water}}|$       (D)  $|\Delta T_{\text{Cu}}| < |\Delta T_{\text{water}}|$

Work:

## Practice Question – Chapter 6 (Heat and Enthalpy)

- 14B.** What is the standard molar enthalpy of combustion of benzene (C<sub>6</sub>H<sub>6</sub>(l))?



| $\Delta H_f^\circ$ / kJ·mol <sup>-1</sup> |        |
|-------------------------------------------|--------|
| C <sub>6</sub> H <sub>6</sub> (l)         | 49.2   |
| CO <sub>2</sub> (g)                       | -393.5 |
| H <sub>2</sub> O(l)                       | -285.8 |

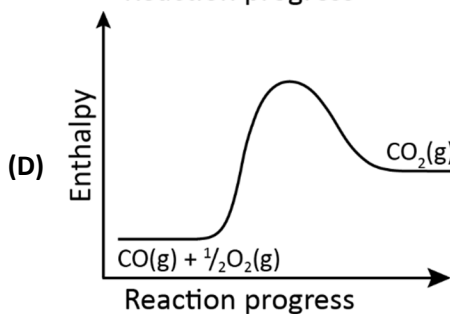
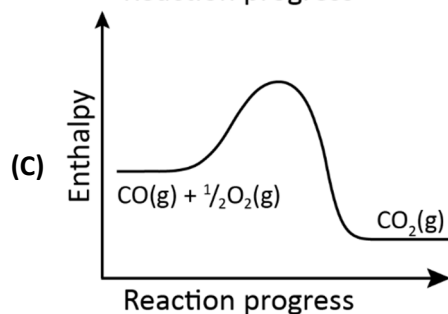
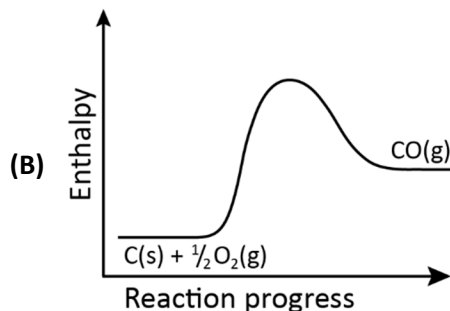
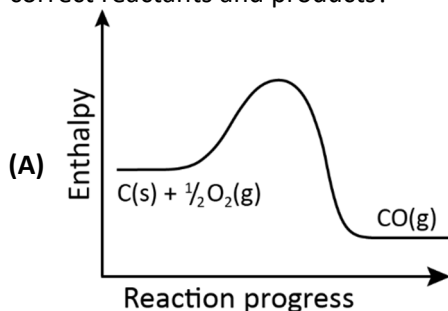
- (A) -6535.2 kJ·mol<sup>-1</sup>      (B) -3267.6 kJ·mol<sup>-1</sup>      (C) -728.5 kJ·mol<sup>-1</sup>      (D) -630.1 kJ·mol<sup>-1</sup>

Work:

For more practice, please see: Study Questions: SQ2-5, 9 (p. 69-75)  
Practice Questions: PQ3-12, 23-30 (p. 75-79)

Study Question – Chapter 6 (Heat and Enthalpy)

**15A.** The enthalpy of formation of CO(g) is  $-110.5 \text{ kJ}\cdot\text{mol}^{-1}$ . Which energy diagram shows this with the correct reactants and products?



Work:

Practice Question – Chapter 6 (Heat and Enthalpy)

**15B.** What is the standard enthalpy of vaporization of water given the two thermochemical equations provided?

|                                                                                   |                    |
|-----------------------------------------------------------------------------------|--------------------|
| $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(g)}$ | $\Delta H_1^\circ$ |
| $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$ | $\Delta H_2^\circ$ |

(A)  $\Delta H_1^\circ - \Delta H_2^\circ$

(B)  $\Delta H_2^\circ - \Delta H_1^\circ$

(C)  $\frac{1}{2}(\Delta H_1^\circ - \Delta H_2^\circ)$

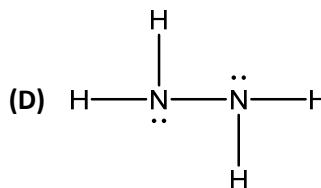
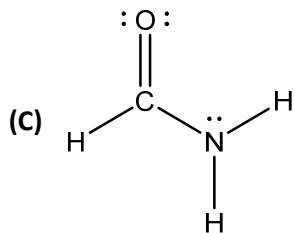
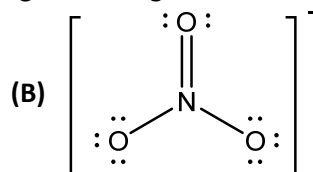
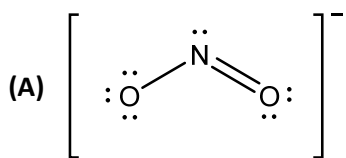
(D)  $\frac{1}{2}(\Delta H_2^\circ - \Delta H_1^\circ)$

Work:

For more practice, please see: Study Questions: SQ6-8 (p. 69-75)  
Practice Questions: PQ13-22 (p. 75-79)

Study Question – Chapter 7 (Structure and Bonding)

**16A.** Which molecule or ion has a non-zero formal charge on nitrogen?



Work:

Practice Question – Chapter 7 (Structure and Bonding)

**16B.** Which compound has the highest lattice energy?

(A) CaO

(B) KCl

(C) MgO

(D) NaCl

Work:

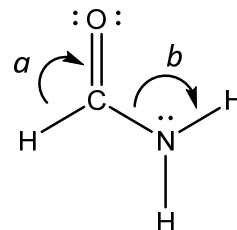
For more practice, please see: Study Questions: SQ1-5 (p. 82-91)  
Practice Questions: PQ1-12, 15-16 (p. 91-95)



Study Question – Chapter 7 (Structure and Bonding)

**17A.** What are the two marked bond angles in the structure shown?

- (A)  $a = 120^\circ$  and  $b = 120^\circ$
- (B)  $a = 109.5^\circ$  and  $b = 120^\circ$
- (C)  $a = 120^\circ$  and  $b = 109.5^\circ$
- (D)  $a = 109.5^\circ$  and  $b = 109.5^\circ$



Work:

Practice Question – Chapter 7 (Structure and Bonding)

**17B.** What is the shape of  $\text{SOCl}_2$ ?

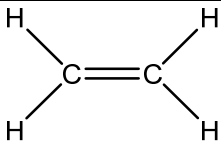
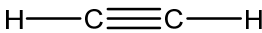
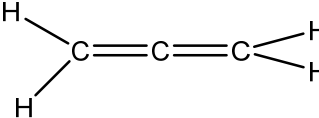
- |                     |                        |
|---------------------|------------------------|
| (A) t-shaped        | (B) tetrahedral        |
| (C) trigonal planar | (D) trigonal pyramidal |

Work:

For more practice, please see: Study Questions: SQ8-9 (p. 82-91)  
Practice Questions: PQ17-20 (p. 91-95)

## Study Question – Chapter 7 (Structure and Bonding)

**18A.** What is the order of least to most sigma bonds for the molecules in the table?

|                                                                                                                         |                                                                                                                          |                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|  <p style="text-align: center;">I.</p> |  <p style="text-align: center;">II.</p> |  <p style="text-align: center;">III.</p> |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|

(A) I. < II. < III.

(B) I. = III. < II.

(C) II. < I. = III.

(D) II. < I. < III.

Work:

## Practice Question – Chapter 7 (Structure and Bonding)

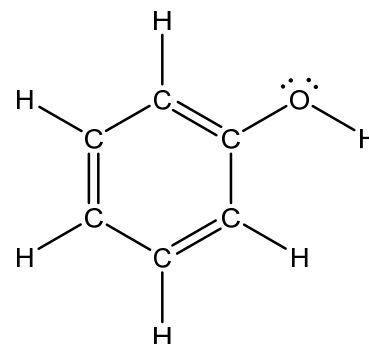
**18B.** What is the hybridization on any carbon atom and the oxygen atom in phenol (shown)?

(A) C = sp and O = sp<sup>2</sup>

(B) C = sp and O = sp<sup>3</sup>

(C) C = sp<sup>2</sup> and O = sp<sup>2</sup>

(D) C = sp<sup>2</sup> and O = sp<sup>3</sup>



Work:

For more practice, please see: Study Questions: SQ11-12 (p. 82-91)  
Practice Questions: PQ24-28 (p. 91-95)

Study Question – Chapter 8 (States of Matter)

**19A.** What is the density of argon at 125°C and 2.0 atm of pressure?

| molar mass / $\text{g}\cdot\text{mol}^{-1}$ |       |
|---------------------------------------------|-------|
| Ar                                          | 39.95 |

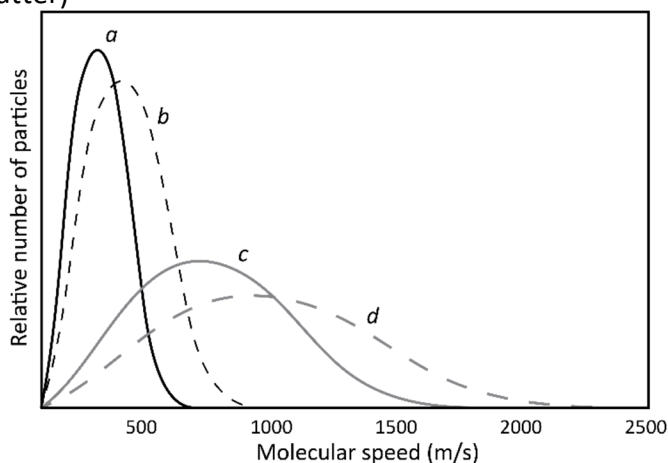
- (A)  $0.61 \text{ g}\cdot\text{L}^{-1}$       (B)  $1.8 \text{ g}\cdot\text{L}^{-1}$       (C)  $2.5 \text{ g}\cdot\text{L}^{-1}$       (D)  $7.8 \text{ g}\cdot\text{L}^{-1}$

Work:

Practice Question – Chapter 8 (States of Matter)

**19B.** For the gases plotted in the figure, what is known?

- (A) Gas *b* is hotter than gas *a*.  
 (B) Gas *c* is lighter than gas *d*.  
 (C) If all of the gases are the same gas, gas *d* is the coldest.  
 (D) If all of the gases are at the same temperature, gas *a* is the heaviest.

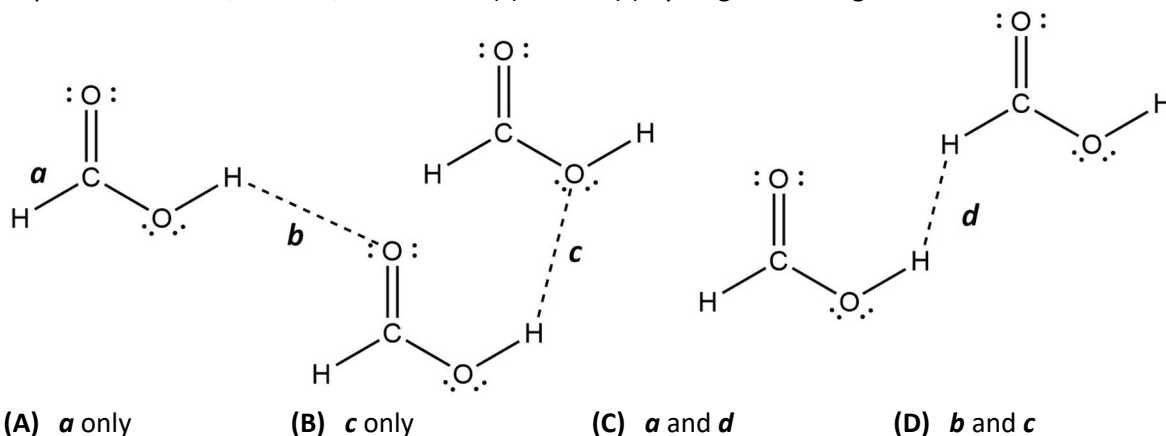


Work:

For more practice, please see: Study Questions: SQ1-6 (p. 98-104)  
 Practice Questions: PQ1-17 (p. 104-108)

Study Question – Chapter 8 (States of Matter)

20A. In pure formic acid,  $\text{HCOOH}$ , which letter(s) denote(s) hydrogen bonding?



Work:

Practice Question – Chapter 8 (States of Matter)

20B. Between  $\text{NF}_3$  and  $\text{NCl}_3$ , which substance has the highest boiling point and why?

- (A)  $\text{NCl}_3$ , because it is polar and  $\text{NF}_3$  isn't.  
 (B)  $\text{NF}_3$ , because it is polar and  $\text{NCl}_3$  isn't.  
 (C)  $\text{NCl}_3$ , because it has more electrons than  $\text{NF}_3$ .  
 (D)  $\text{NF}_3$ , because it can participate in hydrogen bonding and  $\text{NCl}_3$  can't.

Work:

For more practice, please see: Study Questions: SQ7-9 (p. 98-104)  
 Practice Questions: PQ18-25 (p. 104-108)

Good luck on your final exam!

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